

SPL1 Project Proposal 2025

PackAnalyzer

A GUI-Based Network Packet Analysis Tool

Capture · Decode · Inspect · Protect

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PROJECT OVERVIEW

What is PackAnalyzer?

PackAnalyzer is an educational GUI-based network packet analysis tool developed in C using the libpcap library.

The project simplifies the learning process of network traffic capture and analysis by providing an accessible alternative to complex tools like Wireshark.

Goal

Provide students and researchers with a lightweight, visual tool for understanding how network protocols work without the complexity of enterprise tools.

Tech Stack

Language

C (Primary)

Library

libpcap

Build

Makefile

Reference

Wireshark

Packet Capture Engine

Real-Time Capture

- Captures live packets from user-specified interfaces (Ethernet, Wi-Fi)
- Operates in promiscuous mode for comprehensive traffic analysis

Offline Analysis

- Reads and parses pre-captured .pcap and .pcapng files
- Enables retrospective analysis without live network access

Protocol Decoding

Layered OSI Model Interpretation

Layer 2 – Ethernet

Source/Destination MAC, EtherType

Layer 3 – IP

IPv4/IPv6 headers, TTL, Protocol IDs, Src/Dst IP

Layer 4 – TCP/UDP

Source/Destination ports, transport insights

Data Display

Structured Packet View

- Interactive graphical display of decoded headers
- Color-coded protocol tags for quick identification
- Expandable sections for layer-by-layer inspection

Payload Visualization

- Dual-format HEX + ASCII payload viewer
- Deep raw data content inspection for debugging and analysis

Graphical User Interface



Live Dashboard

Real-time packet capture display with live stats



Table View

Timestamp · Protocol · Source · Destination · Length



Detailed Header View

Drill-down into Ethernet, IP, TCP/UDP headers



Payload Inspector

Side-by-side HEX & ASCII payload panel

Deep Packet Inspection (DPI)

TLS/HTTPS Server Name Indication (SNI) Extraction

The Scenario

HTTPS traffic is encrypted-the payload is unreadable. Current tools only show the protocol label HTTPS but cannot reveal which websites are being visited (just raw CDN IP addresses).

The Solution

Implement a lightweight TLS parser to extract the SNI field from the Client Hello handshake — which is transmitted in plaintext before encryption begins.

How It Works

- 1 Client initiates TLS handshake → sends Client Hello
- 2 Parser intercepts the plaintext handshake packet
- 3 SNI field is extracted → domain name decoded
- 4 GUI displays: `www.youtube.com` instead of `142.250.x.x`

High-Speed Signature Detection

Aho-Corasick String Matching Algorithm

The Problem

Scanning each packet against 1,000+ threat signatures using standard string search (strstr) runs at $O(n \times m)$ time complexity, completely freezing the capture engine under real traffic load.

Complexity Gain

strstr / naive
 $O(n \times m)$
SLOW



Aho-Corasick
 $O(n)$
FAST

Algorithm Details

Trie Construction

Builds a finite state machine from all threat keywords (passwords, malware signatures)

Single-Pass Search

Each incoming packet payload traverses the FSM once — finds ALL matches simultaneously

GUI Integration

Threat Signatures tab with real-time pipe alerts:
GUI_ALERT|SIGNATURE_MATCH|
FTP_PASSWORD|192.168.1.5

Thank You Everyone.